More Layers Do Not Mean More Benefit

Geist Atlas · Module 12 · Meeting the Learner

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> [**SUPPORTED**] Supported by the current evidence · **Family:** Meeting the Learner
> **Claim:** A three-part tutor — learner-facing stance, conversation memory, and speaking-style routing — matches the full stack on every measured surface. Stacking does not hurt, but it does not add benefit either. Repeated instructions converge; genuinely new, checkable information is what accumulates.
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> A pre-registered comparison of a small kernel and the full mechanism stack across resistance, suggestion, and trap suites, at n=6 per cell-scenario. It explains why several good instructions often substitute for one another instead of adding up.
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> *Derived view of `paper-full-2.0.md` v3.0.300 — §6.14, §7.11. The paper is canonical; edit it there, not here.*

6.14 Composition: One Tutor from the Validated Mechanisms

Every mechanism this paper validates was tested in its own cell family on its own runtime: calibration on the dialogue engine (§6.1), superego error correction in the multi-agent cells (§6.2), Writing Pad accumulation in the longitudinal arm (§6.6.11), typed action contracts on the LangGraph runner (§6.12), engagement-register routing on the id-director path (§6.7), and pacing guards on the derivation stage (§6.13). No cell composed them. That gap is itself the last untested claim in the mechanism story, and the paper's own findings make a directional prediction about it: universal substitution (§6.4) says calibration renders the superego near-redundant on strong models; the prosthesis-straitjacket result (§7.8.3) says stacked scaffolding can harm capable models; the state-richness reversal (§6.8.6) says added

*This sentence is the only one actually authored by Liam Magee in this paper.

channels that re-encode what the model already infers buy nothing. Composition should therefore be **sub-additive**: a small kernel should match the full stack. This section reports the pre-registered test of that prediction.

Design. Mechanisms were made composable as declared modules in a registry (`config/tutor-blueprint.yaml`): each module records its evidence pointer, the profile factor flags it resolves to, and its portability status on the chosen chassis (the standard runner's id-director path, which already hosts recognition orientation, register routing, and per-turn persona authoring). The registry is enforced-as-check — a test asserts each blueprint cell's explicit YAML factors match the registry resolution — rather than runtime indirection. Two structural findings from the composability audit are recorded in the registry rather than papered over: the conventional superego is *structurally excluded* from this chassis (the id-director re-uses the `superego:` configuration slot as the id and short-circuits the deliberation loop), and the §6.13 pacing guard is not portable (it requires machine-checkable task state these scenario families lack). The Plan 2.x action-contract cycle, by contrast, proved cleanly extractable: the LangGraph nodes are thin wrappers over pure functions, so the select→contract→verify→close loop was ported as per-turn middleware (`services/blueprintActionContracts.js`), with the contract injected into the id's authoring prompt and realization checks recorded in the trace but never used to rewrite the tutor's message. Two cells implement the composition: **cell_199** (kernel: recognition orientation + drama-pad memory + the cell-193 register-router composite) and **cell_200** (kernel + action contracts). The dynamic ego-superego learner is pinned to the base (non-recognition) profile across all arms, registry-enforced, so the simulated-learner instrument is constant. One memory caveat is recorded explicitly: only the interaction-engine drama pad is live on this chassis; the three-layer SQLite pad carrying the §6.6.11 cross-session evidence is bypassed, so that claim does not transfer to the blueprint cells.

Gate (pre-registered, single repeats). The composition gate ran the arms on a Codex-only generation stack (`codex.gpt-5.5` for ego, id, and learner — the stack the cell-193 comparator evidence was generated on) with a Codex CLI v2.2 judge, across three scenario families. On the five gated resistance-breakthrough scenarios (canary `eval-2026-07-02-b0be0524`; Block R `eval-2026-07-02-b4ccb58d`, 20/20 rows), whole-dialogue v2.2 was a flat near-ceiling band — static floor 90.4, cell-193 comparator 91.6, kernel 92.3, full stack 92.0 — supporting sub-additivity and showing no straitjacket regression. On two multi-turn suggestion scenarios (`eval-2026-07-02-fe9404d2`), kernel and full were again indistinguishable (97.2 vs 97.4); the comparator arms (`eval-2026-07-02-a4260aa3`) could not run on the Codex stack because the CLI provider bridge does not reach tutor-core's dialogue engine, so their far lower scores on the canonical OpenRouter stack (cell_5 26.9, budget 9.8) are generation-model-dominated and are *not* composition evidence. A trap-suite block (`eval-2026-07-02-6547750c`) proved structurally non-evaluable: the id-director adapter's traces carry no `policyAction` slot, so the §6.8 binary shift scorer returns no evaluable rows; the block is recorded as instrumentation, not evidence.

The one surprise, run to ground. On the gate's *local* decision metric — the §6.7 rule of gated resistant precondition → route hit → immediate learner uptake — the composite arms trailed the plain cell-193 backbone (2/5 and 1/5 positive vs 5/5) despite routing correctly. Since the kernel differs from cell 193 by exactly one factor (recognition orientation), this raised the possibility that the paper's strongest positive ingredient carries a localized cost in resistance repair. Two further pre-registered runs settled it. A repeats contrast (`eval-2026-07-02-ad0b5a8b`, 30/30 rows) landed in the frozen rule's middle zone; a decision run (`eval-2026-07-03-f877e477` plus one-row supplement `eval-2026-07-03-4deae92a`, 20 rows) completed the corpus at $n = 6$ per cell $\times$ scenario, 30 rows per arm. Final: strict candidate breakthroughs **18/30 vs 17/30** (statistically identical); positive local outcomes **30/30 vs 24/30** — a gap of exactly 6/30, landing on the pre-registered dissolve boundary (real required $\geq$ 9/30). **Verdict: dissolved.** The material claim that recognition orientation impairs local resistance handling is not licensed. Neither is "no cost": the residual is one-directional throughout (cell 193's mean local score is higher on all five scenarios in every accumulation of the data) and concentrated in the frustration subtype, where the kernel converts a still-frustrated learner into productive carry-through work half the time (3/6) against the backbone's 6/6. Whole-dialogue v2.2 is blind to all of this in both directions (repeats covariate 92.2 vs 90.8; decision-run covariate 91.5 vs 91.0 with the kernel's first-turn mean nominally *higher*), reinforcing the §6.7 lesson that register-level effects live below the resolution of transcript-level scoring. A pre-declared stop rule bars further repeats on this contrast.

Reading. The composition prediction is confirmed on every measurable surface: the full stack never beats the kernel, the action-contract module adds nothing detectable at this sample size, and stacking does not harm. Combined with §6.4 and §6.8.6, the practical design consequence is that the "ideal tutor" this paper's evidence licenses is a *small kernel* — orientation, within-dialogue memory, and register routing — not an accumulation of mechanisms; additional machinery should be added only against a demonstrated failure, per the §6.13 pattern. The dissolved-at-boundary frustration residual is retained as a caveat and a target: if the local cost of recognition orientation is real, it is small, subtype-specific, and precisely where the register router already knows the dialogue state — making a scoped suppression (orientation off inside resistance-repair turns) the natural candidate should the question reopen.

Status and caveats (per §5.12.6). Pre-registered design, gate, thresholds, and stop rules are in `notes/2026-07-02-blueprint-composition-plan.md` (§§5, 8, 10, 11), frozen before each paid run. All generation is simulated-learner, single stack (Codex-only), single v2.2 judge (Codex CLI), $n = 6$ per cell $\times$ scenario on the primary contrast; the suggestion-suite comparator arms are stack-confounded as stated; the trap block is non-evaluable; no human-learning claim follows. The decision metric is the judge-free local reporter (`scripts/report-charisma-desire-breakthrough-matrix.js`, arms `blueprint_kernel`/`blueprint_full`); v2.2 scores are covariates only. The positive-gap verdict landed exactly on the dissolve boundary — one row in either direction would have produced "unresolved" (never "real") — and is reported as such rather than rounded in either direction. Total non-subscription cost $\approx$ \$0.30. Artifacts: `config/tutor-blueprint.yaml`, `services/tutorBlueprint.js`, `services/blueprintActionContracts.js`, cells 199–200 in `config/tutor-agents.yaml`, and `exports/charisma-desire-breakthrough-matrix-summary.md`.

7.11 Why composition is sub-additive: instructions converge, signal accumulates

The §6.14 composition result — a three-mechanism kernel matches the full stack, and both merely match strong single-mechanism cells — invites a unifying interpretation of results that this paper has so far reported separately. We offer it here as interpretation, not as a new empirical claim; every observation it draws on is established in the sections cited.

The convergence pattern. Four independent lines of evidence show prompt- and architecture-level mechanisms *substituting* for one another rather than adding. Universal substitution (§6.4): under recognition orientation the superego finds little left to catch, because calibration has already suppressed the outputs it existed to veto. The state-richness reversal (§6.8.6): a richer externalised learner profile does not improve, and can degrade, strategy selection. The orientation-family result (§7.9): matched-specificity prompts from any Hegelian-descendant tradition reproduce recognition's effect within $|d| < 0.2$ — the mechanisms are interchangeable *encodings*, not separate causes. And now composition (§6.14): stacking the validated mechanisms adds nothing the kernel lacks. A fifth, negative line is consistent: the programme's repeated theory-of-mind-redundancy nulls (§6.8.5, the ontology-ToM layer, the stall-watcher precondition null in §6.13.6) all found that channels which re-encode information the model already infers from the dialogue buy no measurable behaviour. A sixth line now extends the law to utterance grain: the first-draft typed-performance boundary (§6.18) found that form made checkable migrates into the harness, harness-owned form does not generalize across worlds, and the repair machinery that enforces form was dispreferred to the raw drafts it replaces in the series' blinded (single-reviewer, exploratory) audit — substitution operating *within* a single turn.

The interpretation. All of these mechanisms act through a single bottleneck: they shape what the generation model attends to in a context it can already see. A sufficiently capable model infers the learner's state, the appropriate stance, and the pedagogically sound move from the raw dialogue; each mechanism contributes only the increment the model would not have inferred unaided, and those increments overlap almost completely because every mechanism is aimed at the same failure mode — generic, transmission-style, learner-blind output. On this reading the mechanisms are surface projections of one latent variable, the intersubjective orientation §7.9 isolates, and a switch already flipped cannot be flipped again. The reading also bears on an anomaly §9 records: the near-identical recognition effect magnitude across structurally different generation models (d = 1.84–1.92) is what one would expect if a single orientation switch, rather than several independent dials, were being set.

Where addition is real. The exceptions define the rule. The interventions in this programme that kept paying off past the first mechanism all inject *information the model does not have*, with a machine-checkable substrate: the proof-state pacing guards of §6.13 (grounded where unguarded tutors fail, precisely because proof-distance is not inferable from the visible page in adversarial geometries), the resistance-signal gates of §6.7 (which enforce a precondition rather than describe one), and the §6.12 action contracts *as instrumentation* (their audit chain works even where their behavioural lift is bounded). Conversely the §6.14 frustration residual — the one place composition showed even a sub-threshold cost — is what rivalrous prompt space predicts: two good instructions (honour the learner's autonomy; force one concrete next step) competing for the same sentence at the same moment.

The design rule. Stated compactly: *instructions converge on the model's existing competence; only new signal accumulates.* An architecture earns a second mechanism not by adding a second description of good teaching but by adding a channel of evidence the model cannot otherwise access — ground truth, enforced preconditions, checkable task state. This is the §6.14 small-kernel consequence generalised, and it is consistent with every substitution, redundancy, and straitjacket result in the paper. Its limit is stated plainly: it is an interpretive synthesis over one system, one scenario ecology, and simulated learners; the constant-effect-magnitude observation it explains remains, evidentially, an anomaly rather than a confirmation.

Where pairing is real: cues and rights. A late pair of §6.24 results adds the complementary tendency and its discriminator. Twice, two components that each measured zero produced the behaviour together: a typed quiet-state card at a detected moment where neither the timing alone nor the typing alone passed its gate (§6.24, Phase Q — "timing and typing separately necessary"), and the deadline-wager, where the standing licence alone went zero across nine delivery-verified live moments, the delivered card alone went zero across three (its best reply withholding exactly the staked send), and card plus licence produced the move at two of three in both model families. These pairs do not contradict the convergence law, because they are not two descriptions of the same competence: one member says *now* (a cue naming the moment) and the other says *may* (a right carved into a standing rule). The discriminator is where the deficit comes from. The convergence pattern governs deficits native to the model — a second description of good teaching finds the switch already flipped. The pairing pattern governs deficits the harness itself manufactures: the wager was never missing from the model (the frameless control produced it five of six times with no rules at all) — it was suppressed by the same contract sentences that produced the section's zero-leak record. Undoing a self-imposed suppression needs both parts, and zeros multiply: a right with no aim is never exercised, an aim with no right is refused. The design rule gains a codicil: before adding a mechanism for a missing behaviour, first establish whether the harness's own standing rules are the cause — if they are, the remedy is a cue-and-right pair priced against the safety rail it cuts into, not another description of the behaviour. The same result re-prices growth: each licence is a hole in a rail (the leak record survived this one at zero across every licensed bench, but that is a measurement per licence, not a property of licences), so the accumulating harness stays governed by the small-kernel consequence — few standing rules, each exception earned against a priced deficit, and the bare-versus-full comparison re-run as the standing total.

The de-substitution challenge. The synthesis above was challenged on a specific ground: every substitution result it draws on was generated against a learner whose non-discrimination is independently measured rather than assumed — SFS = 0.000 (§8.1), a learner that flips to the correct answer at rate 1.000 whether tutor feedback is targeted, mismatched, or generic. Against a learner who yields to anything, competing strategies cannot help looking interchangeable; sub-additivity could be a fact about the learner's persuadability rather than the strategy space. A DAG-pinned learner was built to close this gap (`notes/2026-07-03-dag-pinned-learner-desubstitution-plan.md`): a formal belief-DAG interior with one designated blocking element and a declared desire set, moved from resistant to engaged only by a deterministic check that the tutor's turn addresses the specific blocking element. The learner is held in character by a criterial learner-superego drift gate (extending `services/resistanceSignalGate.js`) that rejects and regenerates any draft turn yielding without the content condition being met, satisfying an undeclared desire, or dropping the resistance early; false-yield to mismatched or generic feedback measured 0.00 throughout instrument validation. Two frozen contrasts followed: whether the multi-strategy router (cell 193) grounds more often than the fixed-strategy floor (cell 186), and, symmetrically, whether the kernel (cell 199) differs from the router.

The confirmatory result. An initial 60-row matrix across the three arms and five resistance subtypes (`eval-2026-07-03-a3cfbe14`) hit 38% drift-gate exhaustion and froze both contrasts as instrument failure — the RLHF pull toward agreeable completion compounded turn over turn faster than a fixed-budget rejection gate could suppress it. A repaired instrument — a turn-decaying character contract, a tutor-visible key, and a semantic release classifier judging whether the tutor substantively supplied the withheld premise — cut exhaustion to 3.3% and surfaced a directional lead (router 4/20 grounded vs. floor 0/20, `eval-2026-07-04-c689cf3a`) one row short of the pre-registered "real" bar. A confirmatory pre-registration (`notes/2026-07-04-desubstitution-confirmatory-prereg.md`) doubled the sample under a further-repaired instrument (exhaustion 0.83%): 120 rows, three arms $\times$ five scenarios $\times$ eight repeats, judge-free deterministic grounding as the sole outcome (`eval-2026-07-04-0d59e4c8`). Both contrasts dissolved at gap zero: every arm grounded this learner at 4/40 (kernel: 4/39 usable, one instrument failure), against frozen thresholds of gap $\geq 7/40$ for a real effect and 3/40 for dissolution — the arc's last sanctioned attempt at either question with this instrument. The generating lead did not replicate at twice the sample: thresholds fixed before the data existed read a lead visible at half the sample as noise rather than signal.

The reading. Substitution survives a learner engineered to yield only to genuine content delivery, not merely a non-discriminating one — strengthening, rather than qualifying, the reading above. The remaining untested axis is not a harder individual learner but learner *populations*: the capability and alertness diversity no single pinned instrument, however criterial, can stand in for. A methods finding travels with the result and sharpens the synthetic-learner limitation already recorded at §8.1: per-turn characterological gating does not compound to dialogue scale on its own — only a turn-decaying contract, loosening precisely where cumulative tutor work has occurred, kept exhaustion low enough to read the matrix at all.

Neighbouring modules: Module 1 — Meeting the Learner Raises the Floor; Module 2 — The Inner Critic Catches Real Mistakes; Module 6 — A Strong Voice Can Crowd Out the Learner; Module 7 — A Bigger Learner Model Did Not Help.